



Mac to Windows

Microsoft wants to make it easier for you to migrate from macOS to Windows 10 with a new tool. <http://dgit.in/MacToWin>



Liquid batteries

A new concept of liquid battery technology called "flow" will cool the battery while powering it. <http://dgit.in/FlowBatt>

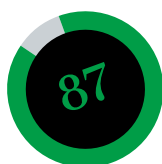
AMD Ryzen 7 1800X

AMD promised to deliver, and deliver they did.



Price
₹37,999

This is the moment that every PC enthusiast has been waiting for. It has been a while since AMD had a product worthy of being pitted against the flagships from Intel so AMD fans had to settle for mid-range SKUs and entry level APUs. Say hello to the AMD Ryzen 7 1800X, an 8 Core / 16 Threaded beast of a CPU that's giving Intel a run for their money, literally. AMD started off the new architecture with a goal of achieving 40 per cent IPC improvement, a goal which they not only achieved but went the extra mile and clocked in an impressive 52 per cent IPC improvement.



Performance..... 84
Value for money..... 90

Specifications

Cores: 8; **Threads:** 16;
Base clock: 3.6 GHz;
Boost clock: 4.0 GHz;
XFR: 4.1 GHz; **Process node:** 14nm

Contact

AMD
Phone: NA
Support: support.amd.com
Website: www.amd.com

a much more granular level of clock speeds. It moves in 25 MHz steps as opposed to the higher 100 MHz steps we've seen so far.

Extended Frequency Range kicks in to provide an additional 100 MHz clock boost taking the Ryzen 7 1800X to 4.1 GHz at peak performance.

Performance

Single threaded applications are where the Kaby Lake Core i7 7700K gets ahead of the Ryzen 7 1800X but the overall margin of difference is within 2-6 per cent. However, with multi-threaded applications, the AMD Ryzen 7 1800X really does a number and races past the 7700K.

Cinebench R15 in single threaded scenario scores 158 on the Ryzen 1800X running at stock, but with a minor overclock on the memory the score rises to 161. Intel Core i7 7700K, however, retains a significant lead at 180. And when you switch to multi-threaded benchmarks, the Ryzen 1800X climbs to a whopping 1638 points and the Intel 7700K only gets to 989. This, is a huge difference which is the result of having twice the number of cores and threads.

PCMark 8 Home when run on accelerated has quite similar scores. However, when we overclock the RAM from 2667 MHz to 3000 MHz, we see an increment of approximately 500 points!

Verdict

AMD's RYZEN marks a new era for computing bringing back competition to a segment that has missed it sorely for quite a few generations. While it may not perform equally in single threaded applications and games, multi threaded applications really excel on the 1800X and that's what the whole industry is moving towards. Right now if you're in the market for a jack of all trades workstation then a Ryzen based system should be your choice as the extra cores and threads will immensely help your productivity. Overall, when compared to the i7 6900K and the i7 5960X, the AMD Ryzen 7 1800X provides way better performance/price. As more software vendors adopt this new architecture and optimise their offerings for Ryzen like they do with Intel, we expect to see this performance value increase even further.

Mithun Mobandas